Assessment Brief

Module title:	Machine Learning
Module code:	COM7022
Assignment title:	COM7022: Assignment
Assignment format:	Portfolio
Word/time limit:	3000 words
File type	.docx / .pdf
Percentage of final grade	This assignment is worth 100% of your final grade for this module.
Submission deadline	See module iLearn page for the date of submission
Grade release	You will normally receive your provisional grade and feedback within 20 working days of the submission deadline

Useful terms:

Learning outcomes (LOs)	The skills and knowledge that you should be able to show in your work.
Marking Matrix	A set of rules or guidelines used to grade or assess work.

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Task summary:

The assignment is given as two tasks. All the tasks are to be completed individually.

- Task 1(Practical Project) will evaluate your practical understand of Machine Learning methodologies for critical analysis, through a real-world Machine Learning project written in Python
- Task 2 (Critical Report) is designed to give you the opportunity to demonstrate your Machine Learning critical reporting skills, comprehensively written in an easily understandable manner, based on your investigation.

A clear, concise analysis for all Tasks is to be given within the submission, complemented with screenshot evidence of all processes and results. You are to submit a single Word or pdf document for all tasks, and your code for Task 1 is to be included within an appendix, so that it can be checked and verified it is working correctly. Your student ID number must be clearly displayed in the uploaded file.

Assignment Details: Task 1: Practical Project

You have recently joined a rapidly growing e-commerce company as a Machine Learning Analyst. The company maintains a large dataset containing product information, pricing details, availability status, sales records, update history, and item locations. Your first task is to develop clustering models using the dataset **Sales-[5444].csv** to group products based on their characteristics and sales performance.

You are required to:

- Prepare the dataset to support effective machine learning analysis.
- Select and justify appropriate features for inclusion within the analysis.
- Apply multiple clustering techniques (e.g., K-Means, Hierarchical Clustering, and DBSCAN) to the dataset.
- Evaluate the suitability of the selected machine learning methods for the problem context.
- Apply data transformation techniques, including scaling, normalisation, and the treatment of categorical variables, where appropriate.

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- Use dimensionality reduction techniques (e.g., Principal Component Analysis (PCA)) to support analysis and visualisation where appropriate.
- Evaluate and compare clustering models using appropriate performance and validation measures.
- Justify the selection of the most suitable model based on technical and business considerations.
- Analyse and describe the characteristics of the resulting clusters and their relevance to the business scenario.

This task will provide the analytical foundation for the executive report presented in Task 2, where findings and recommendations will be communicated to a professional audience.

Task 2: Critical Report:

Drawing on the machine learning analysis undertaken in Task 1, prepare a professional report for an executive audience. The report should:

- Evaluate the suitability of the machine learning methods used to address the business problem, including consideration of feature selection, and model development decisions.
- Compare the performance and effectiveness of the machine learning models applied, critically evaluating their strengths, limitations, and suitability for the given scenario.
- Justify the selection of the final model using both technical and business considerations.
- Present the outcomes in a clear and accessible manner for non-technical stakeholders and include appropriate visualisations, interpretation of key findings, and evidence-based recommendations that support strategic decision-making.

(60 marks, 2000 Words Equivalence, Coded work)
(40 marks, 1000 Words Equivalence, Critical Report)
(LO's: 1, 2. 3)

Note: All analysis must be done in Python, and your notebook should reflect clean code, adequate comments, and informative visualisations.

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Learning outcomes (LO)

By completing this assessment, you will have shown and be assessed on all three learning outcomes:

1. Demonstrate a practical understanding of the characteristics of various Machine Learning methods and their applications in data analysis

2. Select and apply appropriate machine learning techniques for a given scenario

3. Critically evaluate the results from the application of Machine Learning techniques and propose improvements

You will be graded based on how well you meet these learning outcomes. Your marker will use a marking matrix to grade your work, and you can access this by clicking on the submission portal

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